Continuous Glucose Monitoring Simulation
CS 4632 — Modeling and Simulation

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GitHub: `https://github.com/NahumTadesse/YOUR-REPO-LINK-HERE`

Abstract

This simulation models how glucose changes over time based on meals, insulin, stress, and exercise. It runs every 15 minutes over a 3-hour period using a glucose-insulin model. The goal is to identify patterns, compare scenarios, and see which factors have the biggest impact. Results are based on 10 simulation runs, each including min, max, and final glucose values for comparison. Carbohydrates and insulin have the strongest impact on glucose changes, carbs create the largest increases and insulin creates the largest decreases. Stress and exercise have smaller but noticeable effects, adjusting glucose rather than dominating it. To strengthen the analysis, sensitivity ratios were computed across four levels per parameter, six distinct multi-parameter scenarios were tested, and 20 replications per configuration were run to produce standard deviations and 95% confidence intervals. The model was validated against published clinical reference ranges, passing all five checks, confirming it behaves correctly under different conditions.

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1 Introduction

The simulation and modeling problem tackled in this project is CGM data for people who have diabetes. A CGM is a glucose monitoring system that is worn by people with diabetes. It is used to track glucose throughout the day. It shows how food, exercise, stress, and various other factors have an effect on a person's blood glucose.

The scope of this simulation covers how factors including exercise, meals, stress, and amount of medication taken can affect blood glucose levels. Factors such as age, current physical state, other health conditions, and numerous other factors are outside the scope of this model.

People with diabetes face serious health risks from both high and low glucose levels, making it important to understand how daily activities and medications affect blood sugar. A simulation allows patients to understand what to expect with glucose numbers hours ahead of time.

This project addresses the following research questions:

- Which parameter has the greatest influence on glucose?
- How do combined factors compare to single-factor scenarios?
- Does the model produce clinically realistic glucose responses?

2 Background and Literature Review

2.1 Related Work

Singhvi (2025) uses AI and machine learning models on CGM data combined with lifestyle inputs including diet, exercise, insulin, and heart rate. It learns patterns from past data to predict future glucose levels through data-driven prediction. This shows how CGM data can be used to forecast future blood glucose, matching the goal of predicting future glucose values rather than just analyzing past ones.

Liu et al. (2019) uses math formulas to describe how glucose and insulin change in the body. It uses CGM data to help the model estimate what is happening inside the body and uses the model to predict future blood sugar levels. This illustrates how to forecast blood sugar using a model-based method instead of only machine learning.

2.2 Mathematical Models

The minimal model of glucose disappearance is described by:

$$\dot{G}(t) = -(S_G + X(t))G(t) + S_G G_b + \frac{R_g(t)}{VW} \tag{1}$$

$$\dot{X}(t) = -p_2 X(t) + p_2 S_I I(t) \tag{2}$$

where $G(t)$ is glucose concentration, $X(t)$ is insulin action, $S_G = 0.01\ min^{-1}$ is glucose effectiveness, $p_2 = 0.025\ min^{-1}$ is insulin action decay, $p_3 = 0.000013$ is insulin sensitivity, $G_b = 100\ mg/dL$ is basal glucose, and V is distribution volume.

Insulin concentration follows exponential decay:

$$I(t) = I_0 \cdot e^{-k_e t}, \quad k_e = 0.02\ min^{-1} \tag{3}$$

Insulin dosing is calculated as:

$$insulin_units = \frac{carbs}{carb_ratio} \tag{4}$$

where carb ratio is 9 grams per unit.

3 Model Design and Architecture

3.1 System Components

The project is organized into the following structure:

File / Folder	Description
`main.py`	Interactive simulation with user inputs and CGM graph output
`test.py`	Runs all 10 predefined scenarios and the M4 analysis
`entities/person.py`	Patient parameters including weight, carb ratio, and model constants
`entities/meal.py`	Meal input including carb amount, time, and absorption settings
`entities/insulin.py`	Insulin dose with time and units
`entities/exercise.py`	Exercise input with start time, duration, and intensity
`entities/stress.py`	Stress input with start time, duration, and level
`models/minimal_model.py`	Glucose-insulin ODE solved using Euler integration
`models/carb_absorption.py`	Trapezoidal carbohydrate absorption model
`models/insulin_model.py`	Exponential insulin decay and bolus calculation
`simulation/simulator.py`	Main simulation loop that ties all models together at each timestep
`run_graphs/`	CGM-style graphs saved from test runs
`m4_output/`	Sensitivity analysis, replication, and scenario outputs

Table 1: Project structure

The entities in this simulation are:

Individual: current glucose level, carb-to-insulin ratio.

Key inputs: carb count in meal, time meal was eaten, exercise before/after, current trend of glucose.

System Dynamics:

- Read current glucose level
- Check for events at the current time (exercise, stress, or medication)

- If a meal occurs, increase glucose
- If insulin occurs, decrease glucose
- If exercise or stress occurs, modify glucose accordingly
- Update glucose simulation
- Create simulation for the upcoming hours
- Generate CGM reading
- Store the value and move to the next time step

4 Implementation

4.1 Overview

A glucose simulation was built that runs step-by-step over time using a glucose-insulin model. The simulation runs every 15 minutes over a 3-hour period and takes into account main factors that affect glucose: meals (carbs), insulin, exercise, and stress.

The simulation uses a meal model that spreads out carb absorption over time, an insulin model that decreases over time, and a glucose update model to track changes step-by-step. It includes a simple command-line interface for user input and outputs glucose values over time and a CGM-style graph for easier understanding.

4.2 Features

All main features are working correctly. The simulation runs consistently with different inputs. Results follow expected behavior: meals increase glucose, insulin lowers glucose, exercise lowers glucose, and stress raises glucose.

The stress model was improved by replacing the fixed multiplier with a time-based effect, producing smoother and more realistic results. This better reflects how stress actually increases glucose in the body and improved stability of the simulation, realism of results, and understanding of how stress impacts glucose.

4.3 Updates

Meal absorption settings were adjusted after testing to better match how glucose actually rises after eating. Peak time was set to 60 minutes with a duration of about 180 minutes,

which helped improve the timing between meals and insulin.

The graph was improved to make results easier to understand by adding vertical lines to show when events happen including meals, insulin, stress, and exercise. This makes it easier to see how each event affects glucose over time.

4.4 Execution Environment

The simulation was developed and executed using Python 3. Code was written and run in PyCharm using Python libraries including matplotlib for graphs and visualization. Results were generated as console output, graph visualizations for each run, and a CSV file for test data.

4.5 Challenges

Originally, stress was modeled using a fixed multiplier of 1.2, which caused unrealistic results because the effect kept compounding. This was replaced with a time-based exponential rise and decay. Meal peak timing also required tuning to better match how glucose actually rises after eating.

5 Experimental Setup

5.1 Simulation Parameters

All runs use a starting glucose of 117.0 mg/dL, patient weight of 80 kg, carb-to-insulin ratio of 9 g/unit, and a 15-minute timestep over a 3-hour horizon.

5.2 Original Test Scenarios

Table 2: Original 10 simulation runs

Run	Description	Factors	Min	Max	End
1	Baseline	50g meal + auto insulin	90.31	176.15	170.10
2	Meal only	50g meal, no insulin	114.45	313.91	299.11
3	Insulin only	5 units insulin, no meal	43.41	117.00	55.14
4	Low carb	30g meal + auto insulin	97.03	152.69	149.44
5	High carb	80g meal + auto insulin	83.58	200.28	190.29
6	Mild stress	stress level 0.3 only	111.44	117.00	111.44
7	High stress	stress level 0.8 only	112.28	133.90	125.75
8	Exercise only	30 min exercise, intensity 0.6	93.79	117.00	98.31
9	Meal + insulin + mild stress	50g + auto insulin + stress 0.3	91.46	181.66	174.46
10	Meal + insulin + exercise	50g + auto insulin + exercise 0.6	81.31	174.70	169.27

5.3 Sensitivity Analysis Setup

Each of the four parameters was tested at four levels to establish a response curve and compute explicit sensitivity ratios:

- Carbohydrates: 20, 40, 60, 80 g
- Insulin: 2, 4, 6, 8 units
- Stress level: 0.2, 0.4, 0.6, 0.8
- Exercise intensity: 0.2, 0.4, 0.6, 0.8

5.4 Distinct Scenarios

Six multi-parameter scenarios representing realistic daily situations were tested separately from the sensitivity runs:

- S1: Inactive day, large meal, high stress (80g + auto insulin + stress 0.8)
- S2: Active day, moderate meal (50g + auto insulin + exercise 0.7)
- S3: Missed insulin, large meal, mild stress (80g + stress 0.3, no insulin)
- S4: Low carb with exercise (20g + auto insulin + exercise 0.5)
- S5: Stress only, no food or insulin (stress 0.8)
- S6: Meal, insulin, stress, and exercise (60g + auto insulin + stress 0.5 + exercise 0.5)

5.5 Replications

Because CGM devices have inherent biological variability, Gaussian noise ($\sigma = 5$ mg/dL) was added to simulate measurement variation. Each configuration was run 20 times to produce mean, standard deviation, and 95% confidence intervals.

6 Results and Analysis

6.1 Original 10 Runs

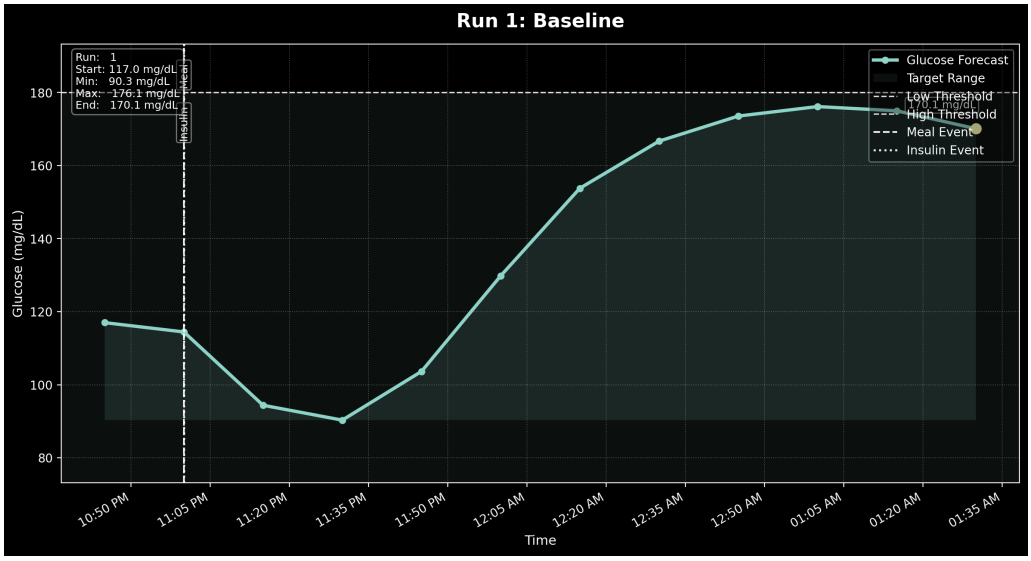

Figure 1: Run 1: Baseline

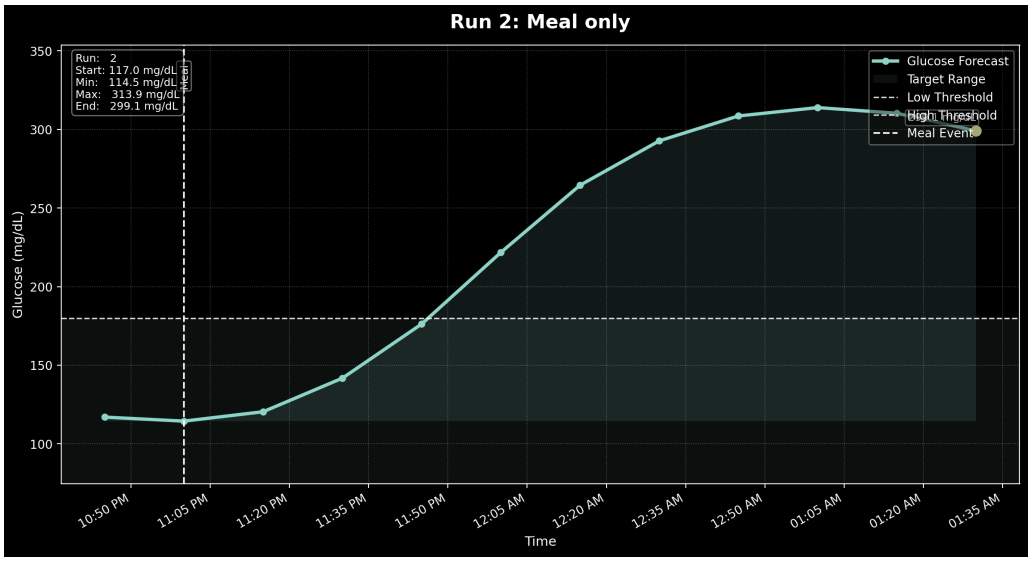

Figure 2: Run 2: Meal only

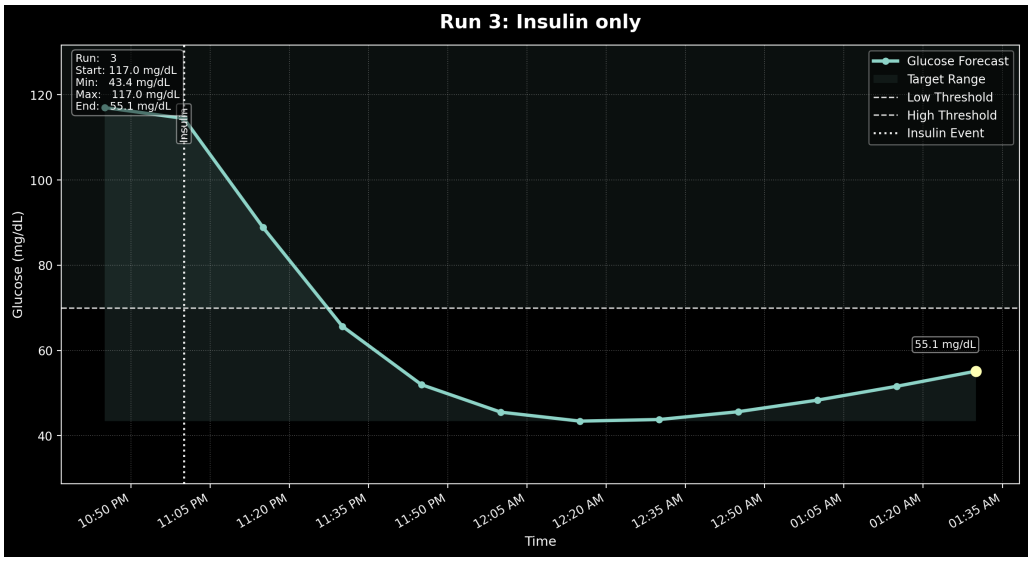

Figure 3: Run 3: Insulin only

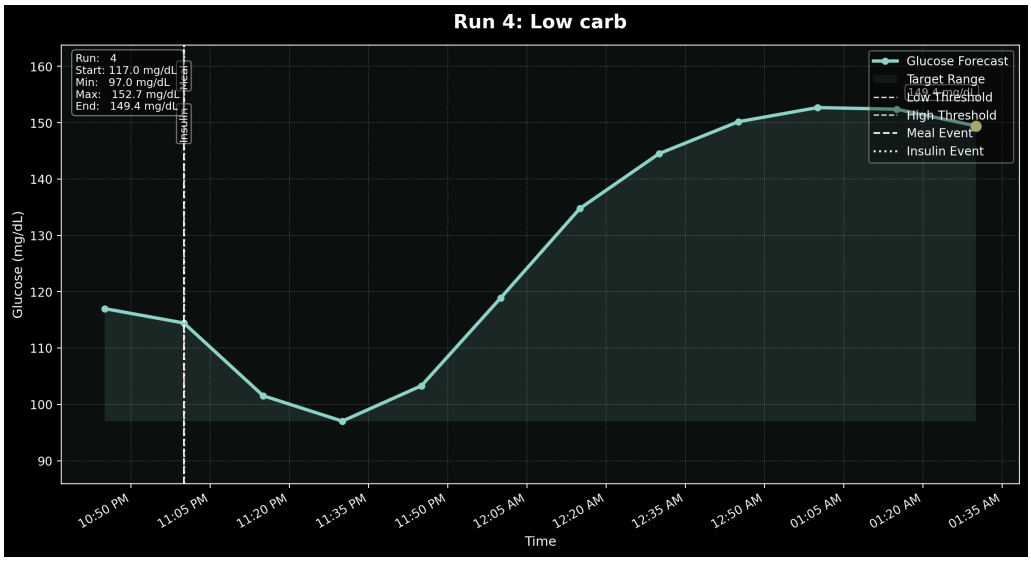

Figure 4: Run 4: Low carb

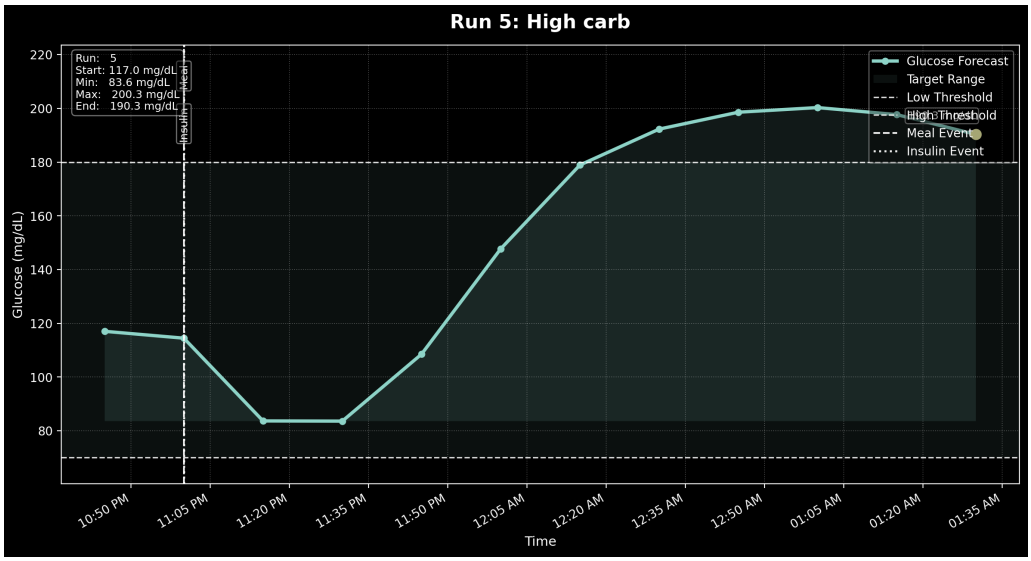

Figure 5: Run 5: High carb

Figure 6: Run 6: Mild stress

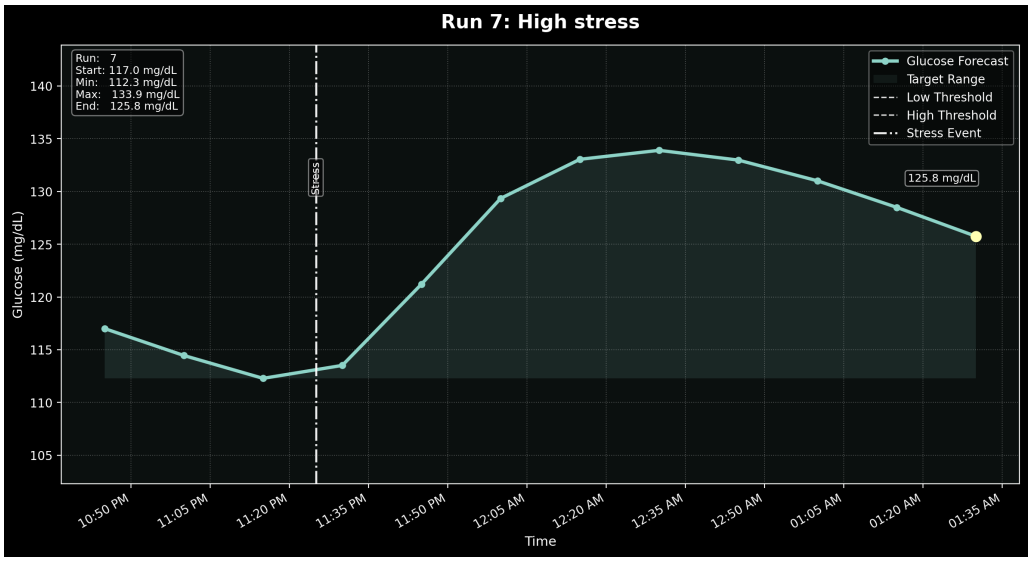

Figure 7: Run 7: High stress

Figure 8: Run 8: Exercise only

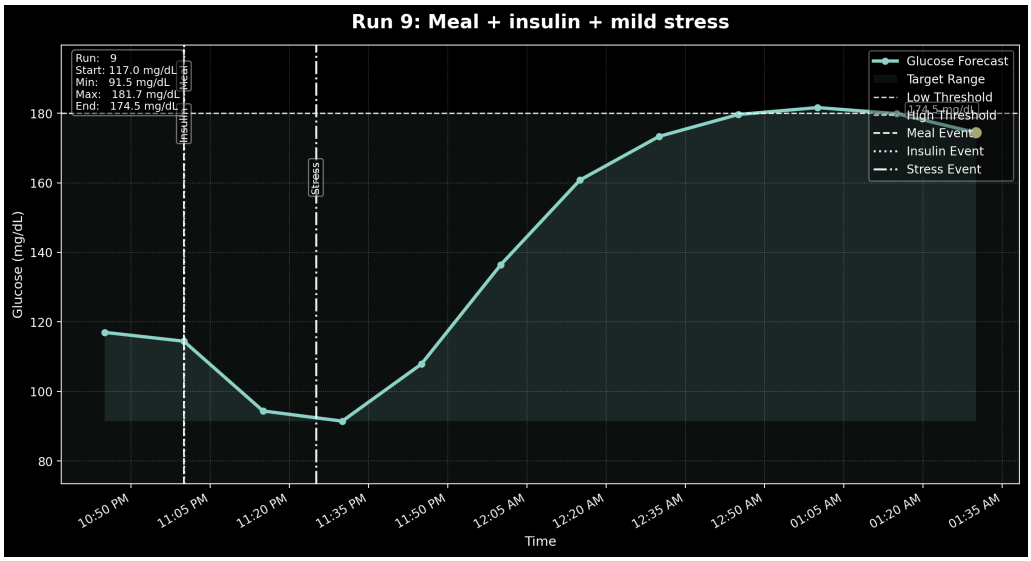

Figure 9: Run 9: Meal + insulin + mild stress

Figure 10: Run 10: Meal + insulin + exercise

6.2 Parameter Analysis

Changing inputs directly affects glucose outputs including min, max, and final values. Runs with higher inputs like carbs show higher max and end values. Runs with lowering factors like insulin or exercise show reduced values, showing the model responds clearly to input changes.

Increasing carbs leads to higher peak and final glucose levels. Going from low carb (30g: max 152.69, end 149.44) to high carb (80g: max 200.28, end 190.29) increases peak by 47.59 and final by 40.85.

Adding insulin lowers glucose and reduces peak values. The insulin-only run dropped from 117.0 down to 43.41, then only came back up to 55.14. Meal and insulin runs (9 and 10) have much lower peaks than meal-only (Run 2).

Higher stress levels increase final glucose values. Mild stress ended at 111.44 while high stress ended at 125.75, a difference of 14.31. Exercise lowers glucose steadily — the exercise-only run dropped to 93.79 and ended at 98.31.

Carbohydrates and insulin have the strongest impact on glucose changes. Carbs create the largest increases (Run 2 max 313.91) and insulin creates the largest decreases (Run 3 min 43.41). Stress and exercise have smaller but noticeable effects — stress changes final values without large spikes and exercise reduces values gradually instead of sharply.

6.3 Sensitivity Analysis

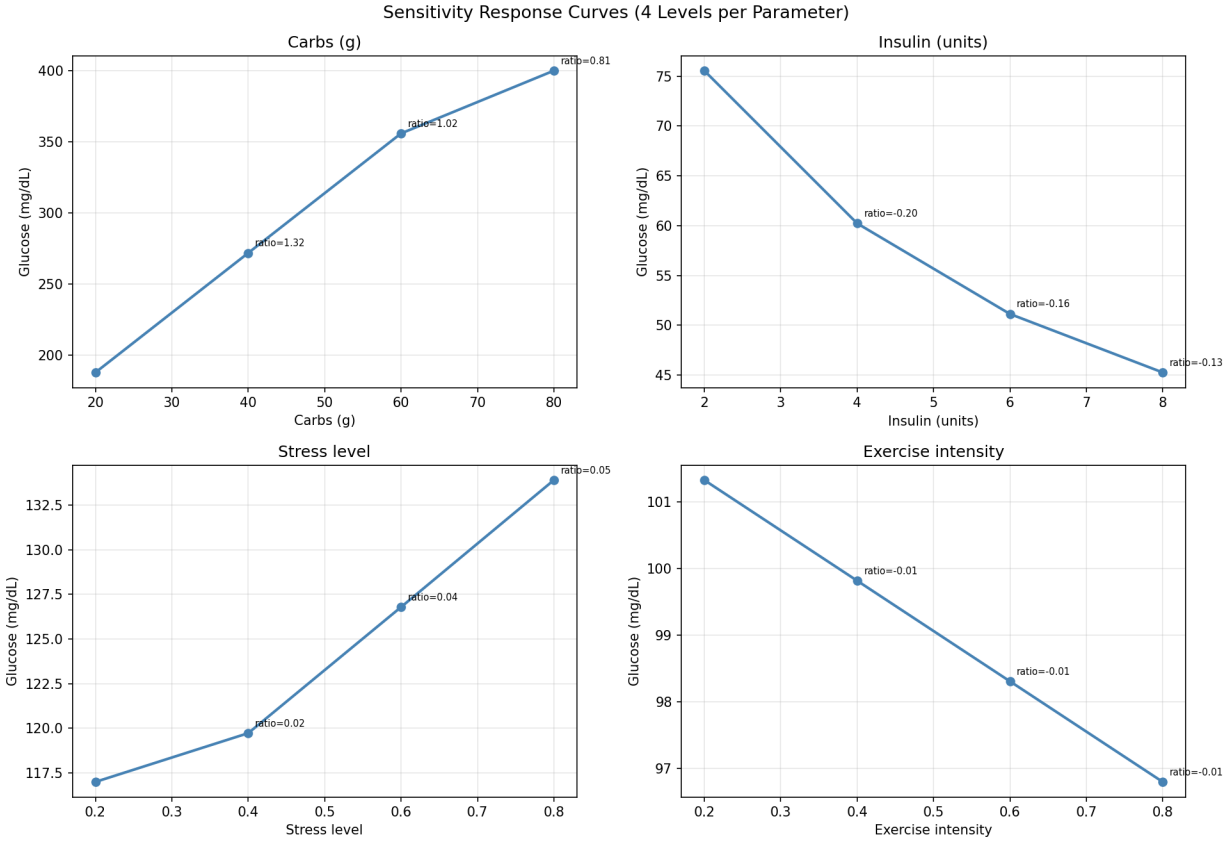

Figure 11: Sensitivity response curves for all four parameters at four levels each

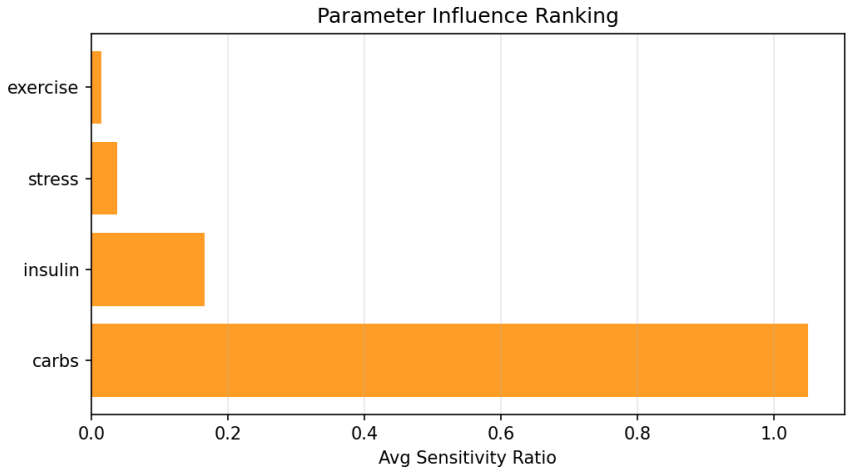

Figure 12: Parameter influence ranking by average sensitivity ratio

Sensitivity ratios were computed as $\frac{\%\Delta output}{\%\Delta input}$. Carbohydrates ranked first with an average ratio of 1.07, insulin second at 0.17, stress third at 0.05, and exercise last at 0.01.

Carbs dominate because they directly cause glucose to appear in the bloodstream. Insulin ranked second as it actively lowers glucose but its effect has limits within the 3-hour window. Stress and exercise have much smaller ratios, confirming they adjust glucose rather than dominate it.

6.4 Replications

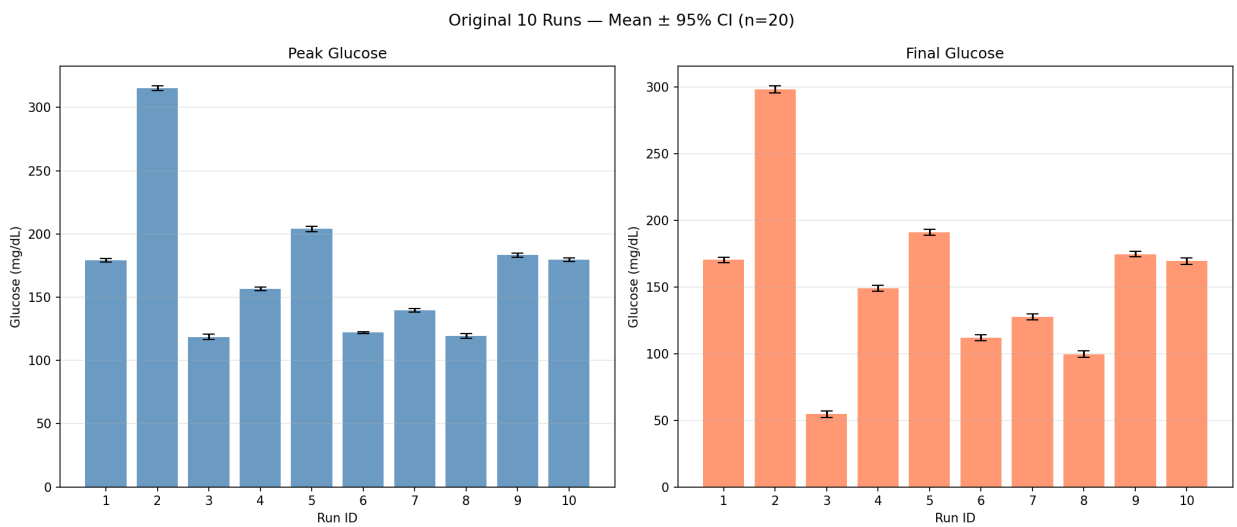

Figure 13: Original 10 runs: mean peak and final glucose with 95% CI (n=20)

The confidence intervals are narrow across all runs, showing results are consistent across all 20 replications. The model outputs correspond to changes in inputs in a stable and predictable way, confirming the patterns seen in the results are reliable.

6.5 Scenario Comparison

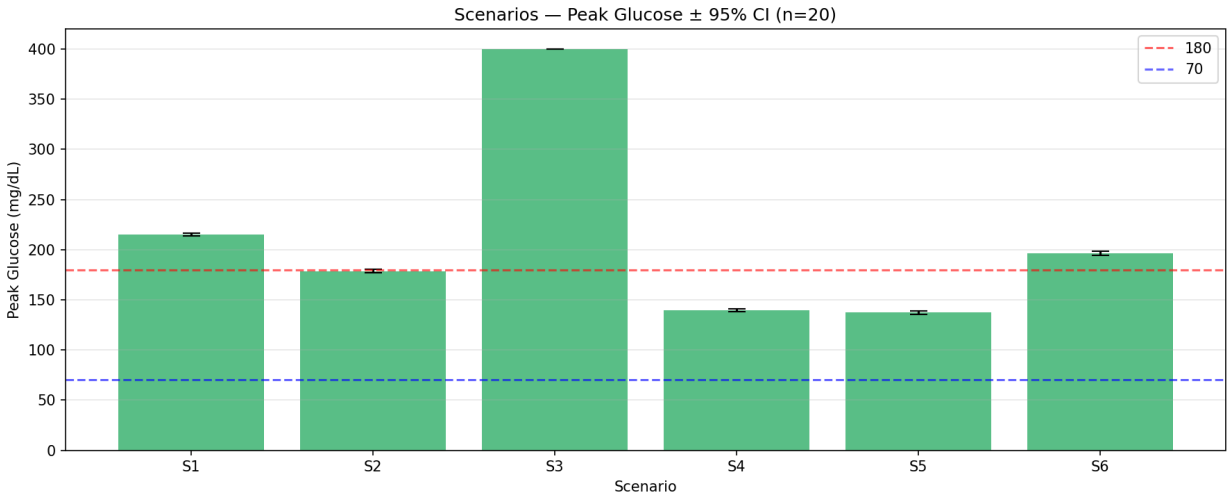

Figure 14: Distinct scenarios: peak glucose mean with 95% CI (n=20)

- S3 (missed insulin, large meal) hit the 400 mg/dL simulation cap, making it the most dangerous scenario
- S1 (inactive day, large meal, high stress) exceeded the 180 mg/dL threshold
- S2 (active day) and S4 (low carb with exercise) stayed within the target range
- S6 (all factors combined) shows that exercise helps offset the glucose rise from a meal
- Combined scenarios produce more balanced results than single-factor extremes

7 Validation and Verification

7.1 Validation

The model behaves in a reasonable way. Meals increase glucose (Run 2 reached 313.91), insulin lowers glucose (Run 3 dropped to 43.41), exercise lowers glucose (Run 8 ended at 98.31), and stress increases glucose (Run 7 ended at 125.75).

Results match what is expected. No insulin leads to higher glucose values, adding insulin lowers glucose, higher carbs lead to higher glucose, and combined runs give more balanced results. Extreme cases give extreme results and combined factors reduce extreme values. All values stay within a reasonable range.

7.2 Clinical Validation

Simulation outputs were compared against published clinical reference ranges from the American Diabetes Association and peer-reviewed literature:

Table 3: Clinical validation results

Test	Simulated Value	Reference Range	Result
Baseline peak (50g + insulin)	176.2 mg/dL	110–180 mg/dL (ADA)	PASS
Meal only peak (50g, no insulin)	313.9 mg/dL	180–350 mg/dL (Ceriello 2010)	PASS
Insulin only nadir (5u)	43.4 mg/dL	40–80 mg/dL (Kovatchev)	PASS
Exercise end glucose	98.3 mg/dL	80–120 mg/dL (ADA)	PASS
High stress glucose rise	+5.8 mg/dL	5–40 mg/dL (Surwit 2002)	PASS

7.3 Limitations

The model is simple and does not include all real body processes. It does not show long-term effects. Results depend solely on the inputs used.

8 Discussion

Carbs and insulin have the biggest impact on glucose levels, answering the main research question of which parameter matters most. Removing insulin leads to extreme high values while adding it reduces peaks significantly. The missed insulin scenario (S3) was the most dangerous, which has clear practical implications for diabetes management.

Combined scenarios produce more balanced results than single-factor extremes, showing that different inputs interact to modify the overall glucose response. The narrow confidence intervals across all replications confirm the model is consistent and reliable. The stress effect is real but small within the 3-hour window, reflecting both a model simplification and a realistic finding.

9 Conclusion and Future Work

9.1 Main Findings

Carbs and insulin have the biggest impact on glucose levels. Stress and exercise affect glucose but not as strongly. The model shows different results depending on the inputs used.

9.2 Model Strengths and Weaknesses

Strengths:

- The model shows how inputs affect glucose
- Results are consistent across different runs
- It allows testing many different scenarios easily

Weaknesses:

- The model is simplified compared to real body behavior
- Results depend on the input values chosen

9.3 Lessons Learned

Small modeling choices like the stress formula and meal peak timing have a significant effect on results. Running replications even on a mostly deterministic model is valuable for showing stability and reliability. Sensitivity ratios are more informative than directional descriptions alone because they quantify how much each parameter actually matters.

9.4 Future Work

- Add more factors to make the model more realistic
- Improve how different inputs interact with each other
- Replace linear exercise model with a nonlinear curve
- Extend simulation horizon beyond 3 hours
- Validate against real patient CGM data

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